Data bibliography / attribution

Orbital Technosphere / Global

01 CelesTrak / orbital elements and memberships

26 OMM CSV catalogs. Collected 2026-11-03T17:13:12Z to 2026-11-03T17:14:05Z.
Active catalog supplies the satellite population; membership catalogs identify named fleets. Three bounded debris catalogs supplement it.
Catalogs: active, starlink, oneweb, kuiper, qianfan, hulianwang, gnss, iridium-next, globalstar, orbcomm, resource, science, stations, fengyun-1c-debris, iridium-33-debris, cosmos-2251-debris, planet, spire, intelsat, ses, eutelsat, telesat, gps-ops, glo-ops, galileo, beidou.
Attribution: CelesTrak. Exact feed URLs, element epochs and hashes are retained in the source ledger.
https://celestrak.org/NORAD/documentation/gp-data-formats.php

02 SGP4 / Vallado, Crawford, Hujsak and Kelso

Positions propagated from the pinned elements using the reference SGP4 implementation, WGS-72 and AFSPC mode.
Calculation instant: 2026-11-03T17:10:00Z. 44 records excluded by epoch; 0 propagation failures.
Public element uncertainty and approximate Earth rotation limit these visualization positions.
https://celestrak.org/publications/AIAA/2006-6753/

03 Natural Earth / terrestrial context

Natural Earth 1:10m physical land provides the subdued geographic background.
Attribution: Natural Earth. Public-domain cartographic data.
https://www.naturalearthdata.com/downloads/10m-physical-vectors/10m-land/

Historical validation / NASA SSCWeb

Reference instant: 2026-08-13T15:36:00Z.
Historical regression reference only. The November refresh was unreachable; this response does not provide an independent comparison of the plotted November positions.
https://sscweb.gsfc.nasa.gov/WebServices/REST/

Cartography / Cartofreako

Projection: Authagraph. The projection bibliography credits its mathematical and design sources.
Separate map and legend composition follows the Setouchi ST2 layout. Typography: Atkinson Hyperlegible; bundled fonts under the SIL Open Font License.
https://bdekoz.github.io/cartofreako/docs/pages/projections/authagraph/bibliography.html

Pinned inputs / source ledger

The profile records the calculation time, observer, filters and every catalog URL. Adjacent acquisition.json and SHA256SUMS record retrievals and original input bytes.
Map and legend carry the same timestamp and counts; the complete acquisition ledger is embedded in this SVG.
Map image scaled to 98.863636% with proportional sheet margins.
https://github.com/bdekoz/cartofreako/blob/main/assets.static/orbital-technosphere/20261103/orbital-technosphere-profile.json

Setouchi AuthaGraph: +4.35-inch printed horizontal move on a 43.5-inch image. Antarctic land is deliberately displaced to the lower left; object coordinates retain source registration.

MSP speculation 4 / Midnight blue / champagne

Randoma11y: ground #010434; marks/type #ead784. sRGB contrast 13.62:1 (screen only).
https://randoma11y.com/%23ead784/%23010434
Epson Stylus Pro 9000 / metallic glossy paper / cyan and light magenta reported low; grays reportedly print violet.
Intended RGB hues; final printed hue is unmeasured. No green correction, ink separation or metallic reflection simulation.

Orbital technosphere
Global / Authagraph

SGP4 positions / 2026-11-03T17:10:00Z

17,909 plotted objects / named fleets and residual roles

Seven-day element age; six-hour future tolerance.

Starlink 11155
OneWeb 651
Kuiper 391
Qianfan 256
Hulianwang 235
Iridium 80
Globalstar 36
ORBCOMM 14
Planet 129
Spire 72
Intelsat 55
SES 67
Eutelsat 29
Telesat 17
GPS 32
GLONASS 24
Galileo 31
BeiDou 52
Debris 1101
Other active 3247
Other navigation 19
Other Earth observation 154
Science 43
Human presence 19

One object per NORAD ID. Overlapping named memberships use profile order.

Starlink NORAD IDs are visible at 7 pt in a separate id layer. All name layers are hidden.

Black x: debris. Sunburst: Starlink. Square: Intelsat. Triangle: remaining Earth observation.

Active-marker opacity records illumination. Debris remains solid black.

Terrestrial subpoints; bounded representative ground tracks; reference site marked.

Starlink sunbursts: 1.5× linear size on both orbital maps.
Starlink text: 7 pt; all satellite markers paint above it.
All terrestrial land, including Antarctica, paints below the data.